

Leakage-Safe Evaluation of Envelope Features for Bearing Fault Detection on an Edge Gateway

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Abstract—Vibration-based bearing fault classifiers often report near-perfect accuracy on public benchmarks, but these figures usually come from splits that place windows of the same physical fault in both training and test data. We evaluate time-domain and envelope features on a ladder of increasingly strict splits on the CWRU and Paderborn bearing datasets. The strictest splits hold out whole faults, severities and damage types. Scores are pooled across folds, with 95% confidence intervals from resampling whole faults. Random-window splits score 1.000 macro-F1 on CWRU. Holding out each physical fault reduces this to 0.457, and to 0.745 on Paderborn. Under the honest splits, envelope features at the kinematic fault frequencies outperform time-domain features: by +0.24 [+0.15, +0.33] on Paderborn leave-one-bearing-out and +0.44 [+0.28, +0.61] on CWRU leave-one-severity-out. The gain depends on knowing the shaft speed. When speed is estimated from the vibration within the rated range, the gain is retained on CWRU but largely lost on Paderborn. On an Orange Pi 5 Plus edge gateway, the envelope pipeline runs 99× faster than real time on one low-power Cortex-A55 core. Across all eight cores the gateway serves about 353 sensors in real time.

Index Terms—bearing fault diagnosis, data leakage, envelope analysis, vibration monitoring, edge computing, machine learning evaluation

I. INTRODUCTION

Rolling-element bearings are a leading cause of rotating-machine failure, and vibration monitoring can detect their faults before breakdown. The Case Western Reserve University (CWRU) data^[1] have become the standard test bed. Reviews of deep-learning methods compile many CWRU accuracies close to 100%^{[2], [3]}.

A classifier is only useful on a bearing it has never seen, however. CWRU records each seeded fault at four motor loads, and a recording is cut into many short windows. When training and test windows are drawn at random, both sides contain windows from the same physical fault, and a model can score highly by recognising that fault rather than learning the fault type. This form of leakage, where test data are not independent of training data, has inflated reported results across many fields^[4]. Benchmark studies of CWRU itself report mislabelled and unusable files and call for stricter evaluation practice^{[5], [6]}.

This paper asks which features still work when the evaluation is honest. We compare broadband time-domain features with envelope features computed at the bearing's kinematic

fault frequencies^[7]. We do so on CWRU and on the Paderborn data^[8], which provide many independent bearings and naturally worn as well as artificially made damage. The contributions are:

- 1) a ladder of split protocols, from random windows to held-out faults, severities, damage types and datasets, with every recording checked to sit on one side of each split;
- 2) pooled, fault-level scoring; a baseline fit on training labels only, and 95% confidence intervals from resampling whole faults;
- 3) evidence that envelope features beat time-domain features on the honest splits, together with an ablation showing how much of that advantage survives when the shaft speed is estimated rather than measured;
- 4) cost and timing on an Orange Pi 5 Plus edge gateway serving many sensors, including a single low-power core.

II. RELATED WORK

Smith and Randall applied established diagnostic techniques to the entire CWRU data set and categorised the files by how clearly they show their labelled fault. Their analysis identified recordings that are mislabelled, clipped or dominated by noise^[5]. Hendriks et al. revisited the use of CWRU for benchmarking machine learning methods^[6]. Zhang et al. and Neupane and Seok surveyed deep-learning methods for bearing diagnostics and compiled their CWRU accuracies^{[2], [3]}. Kapoor and Narayanan surveyed leakage across machine-learning-based science and found it in 17 fields, affecting 294 papers^[4].

Envelope analysis is the classical way to expose bearing faults: band-pass filtering around a structural resonance and demodulating reveals impacts at the fault frequencies^[7]. The spectral kurtosis can choose that band automatically^[9]; we use a fixed band, which suits a fixed sensor installation. The Paderborn data^[8] add many independent bearings with both artificial and real damage. That makes possible the test that matters most for deployment: training on lab-made faults and testing on worn ones.

Table 1

DATASETS. A FAULT IDENTITY IS ONE PHYSICAL DAMAGED BEARING (OR ONE RIG CONFIGURATION).

Dataset	Recordings	Fault ids	Role
CWRU 12 kHz drive end	52	14	main results
Paderborn	2,319	29	main results
MaFaulDa	1,951	24	held-out imbalance severity
MFPT	20	3	cross-dataset transfer

III. DATA

Table 1 summarises the four datasets. CWRU^[1] and Paderborn^[8] carry the main results; MaFaulDa^[10] supplies an imbalance experiment and MFPT^[11] a third dataset for transfer. A *fault identity* is one physical damaged bearing, or one physical configuration on a single rig. Every split keeps each fault identity, and each recording, on one side only.

For CWRU we use the 12 kHz drive-end set and the normal baseline, resampled to 12 kHz: 52 usable recordings, 205 windows of 4 s with 50% overlap, and 14 fault identities. Twelve files are excluded as defective, extending the list of Smith and Randall^[5]. These are clipped or noisy recordings, and files containing no array named after their file number. A notch’s identity ignores motor load, and outer-race notches at different positions are separate faults. CWRU contains a single healthy bearing. For Paderborn we exclude the three bearings with combined inner- and outer-race damage, since they have no single-fault class. That leaves 2,319 recordings of 29 bearings.

IV. METHOD

A. Features

The *time_only* set has 17 features. Ten are time-domain statistics: RMS, peak, peak-to-peak, standard deviation, Pearson kurtosis, skewness, and the crest, shape, impulse and clearance factors. Seven are broadband spectral features: centroid, entropy, flatness, roll-off and band energies. The *envelope_ratio* set (39 features) adds features computed at the fault frequencies: six shaft-order amplitudes, twelve envelope band energies and four sideband energies. A shaft rate f_r and a bearing with n balls of diameter d on pitch diameter D at contact angle φ give the outer- and inner-race fault frequencies

$$f_{\text{BPFO}} = \frac{nf_r}{2} \left(1 - \frac{d}{D} \cos \varphi \right), \quad f_{\text{BPFI}} = \frac{nf_r}{2} \left(1 + \frac{d}{D} \cos \varphi \right). \quad (1)$$

The signal is band-passed to 2–5 kHz, demodulated with the Hilbert transform, and decimated by 16; $E(f)$ is the magnitude spectrum of the resulting envelope. Each envelope feature is the energy within $\pm 2\%$ of a harmonic h of a fault frequency f_0 , normalised by the total envelope energy:

$$x_{f_0,h} = \frac{\sum_{|f-hf_0| \leq 0.02hf_0} E(f)^2}{\sum_f E(f)^2}. \quad (2)$$

Because each feature is a ratio, it is unaffected by the overall signal amplitude.

B. Split protocols

Each protocol holds out a stricter unit than the one before. P0 holds out a random 20% of windows; it leaks by design and is shown only for contrast. P1 holds out one recording, P2 one motor load or operating condition, and P3 one fault identity. P3b, on Paderborn only, trains on artificial damage and tests on real damage. P4 holds out one fault size, and P5 a whole dataset. Every fold is checked so that no recording, and under P3 no fault identity, appears in both training and test data. Splits are frozen as hashed manifests before scoring.

C. Models and scoring

Logistic regression (class-balanced) is the main model, and gradient-boosted trees (GBDT, 100 trees, depth 4) are the second. Standardisation is fit on each training fold only. Leave-one-group-out folds are often small or contain a single class, so per-fold macro-F1 is ill-defined. We therefore pool the test predictions $\hat{y}$ of all folds and score them once:

$$F1_{\text{macro}} = \frac{1}{|C|} \sum_{c \in C} \frac{2TP_c}{2TP_c + FP_c + FN_c}, \quad (3)$$

where C is the set of classes present in the dataset. The baseline predicts the most frequent training-fold label. Confidence intervals are 95% percentiles over 1,000 resamples of fault identities with replacement. The envelope–time difference Δ uses the same resample for both feature sets.

D. Shaft speed

The envelope and order features need f_r . We compare three sources. The first is the speed recorded with the data (*measured*), which requires a tachometer. The second is an estimate from each window alone, which scores candidate rates between 10 and 60 Hz by their first three harmonics against the spectral noise floor. The third is the same estimator searched only within $\pm 10\%$ of the machine’s rated speed. The estimator also reports a confidence: the weighted share of the $1\times$, $2\times$ and $3\times$ harmonics that clear a noise-derived threshold. Below 0.35, the speed-dependent features are zeroed and only speed-free features remain.

V. RESULTS

A. Effect of split strictness

Figure 1 and Table 2 show pooled macro-F1 along the ladder. On CWRU, random windows (P0) score 1.000 for every feature set. Holding out a recording (P1) or a load (P2) still scores above 0.9, because the same physical fault remains in training. Holding out the fault itself (P3) drops *envelope_ratio* to 0.457 and *time_only* to 0.298. On Paderborn the random-window split already separates the feature sets (0.945 vs 0.749). Leave-one-bearing-out gives 0.745 vs 0.506. The train-majority baseline stays near 0.168.

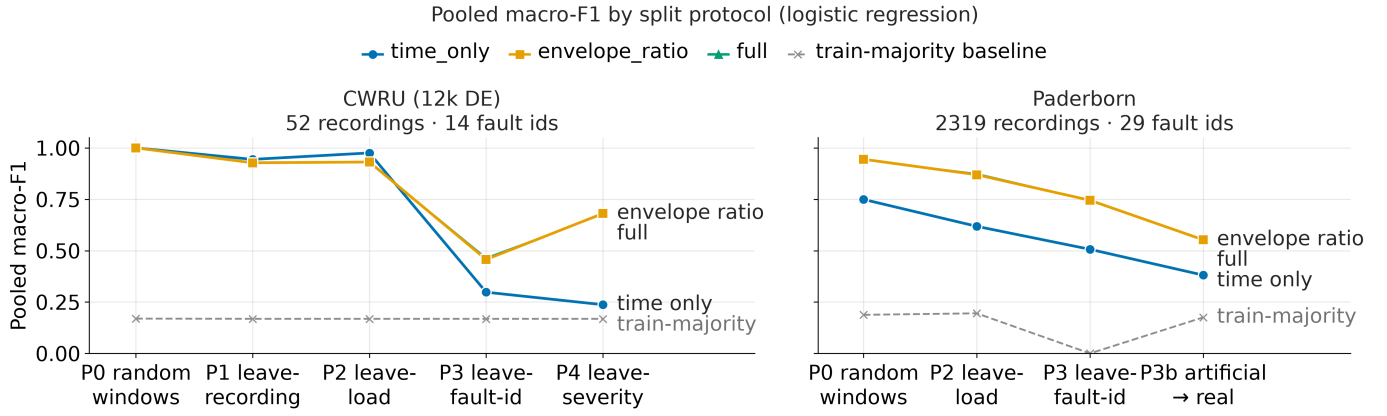


Figure 1. Pooled macro-F1 by split protocol for each feature set on CWRU (left) and Paderborn (right), logistic regression, measured shaft speed. The dashed line is the train-majority baseline.

Table 2
POOLED MACRO-F1 ALONG THE SPLIT LADDER (LOGISTIC REGRESSION, MEASURED SHAFT SPEED). ENV. = ENVELOPE_RATIO, TIME = TIME_ONLY; - = PROTOCOL NOT APPLICABLE TO THE DATASET.

Protocol	CWRU		Paderborn	
	Env.	Time	Env.	Time
P0 random windows	1.000	1.000	0.945	0.749
P1 leave-one-recording-out	0.928	0.944	-	-
P2 leave-one-load-out	0.931	0.975	0.869	0.618
P3 leave-one-fault-out	0.457	0.298	0.745	0.506
P3b artificial → real	-	-	0.553	0.381
P4 leave-one-severity-out	0.681	0.237	-	-

B. Envelope versus time-domain features

Table 3 reports the honest splits with confidence intervals. With logistic regression, envelope features gain +0.24 [+0.15, +0.33] on Paderborn P3, +0.17 [+0.06, +0.30] when training on artificial and testing on real damage, and +0.44 [+0.28, +0.61] on CWRU P4. On CWRU P3 the gain (+0.16 [-0.06, +0.40]) is not significant with only 14 faults. GBDT shows the same collapse from random to honest splits (Figure 2) and confirms the CWRU P4 and Paderborn P3b gains. On Paderborn P3 it extracts more from the time-domain features, and the gap narrows to +0.04 [-0.04, +0.11]. Across 5 GBDT seeds, P3 macro-F1 moves by at most 0.004 (standard deviation).

CWRU’s single healthy bearing makes healthy F1 zero under P3 by construction, because its fold trains with no healthy data. Averaged over the three fault classes only, CWRU P3 scores 0.609 [0.32, 0.82] with envelope features and 0.397 with time-domain features. Figure 3 shows where the errors fall. Figure 4 shows that envelope features classify most held-out outer- and inner-race faults correctly, while ball faults remain partial.

C. Overfitting to fault identity

Figure 5 shows GBDT log-loss after each of 300 boosting stages. Under random windows, held-out loss keeps falling and held-out macro-F1 reaches 1.00 on Paderborn. Under leave-

one-fault-out, held-out loss is lowest after 10 (CWRU) and 22 (Paderborn) stages, then rises while training loss approaches zero. The additional trees therefore memorise the training faults rather than the fault types. The stage of minimum loss is chosen on the held-out data, so it is descriptive, not a tuned stopping point.

D. Transfer and severity

Across datasets (Figure 7), a model trained on Paderborn with envelope features transfers to MFPT (0.81) and CWRU (0.65). The time-only model reaches at most 0.53 off its own dataset. On MaFaulDa, imbalance remains detectable at held-out masses for every feature set: 0.922 (envelope) and 0.951 (time) when interpolating, against a baseline of 0.444. That task is easy enough that it does not separate the feature sets (Figure 6).

E. Physical check

Figure 8 shows real CWRU envelope spectra with the kinematic frequencies of (1) marked. The inner-race fault peaks at BPFI and the outer-race fault at BPFO and its harmonics. The ball fault produces only weak peaks, consistent with its partial classification in Figure 4.

VI. SHAFT SPEED: MEASURED VERSUS ESTIMATED

Table 4 and Figure 9 show that the envelope advantage depends on the accuracy of the shaft speed. On CWRU, the estimator restricted to the rated speed places 100% of windows within 2% of the measured speed. With that prior, the gain equals the measured-speed gain (P4: +0.45 [+0.29, +0.62]). Over the generic 10–60 Hz range, it locks to twice the shaft rate on 67% of windows. The likely cause is the 60 Hz mains line, which lies close to $2 \times$ the 29.95 Hz shaft rate.

On Paderborn the shaft line is weak. The estimator’s confidence is usually low, so only 40% of windows keep the speed-dependent features, and the gain shrinks to +0.05 [-0.02, +0.12]. The estimator was fixed against unit tests on synthetic signals; its real-data accuracy was measured once, after those tests passed.

Table 3

POOLED MACRO-F1 ON THE HONEST SPLITS WITH 95% FAULT-LEVEL BOOTSTRAP INTERVALS, MEASURED SHAFT SPEED. Δ IS ENVELOPE_RATIO MINUS TIME_ONLY FROM A PAIRED BOOTSTRAP; * MARKS INTERVALS THAT EXCLUDE ZERO.

Split	Model	envelope_ratio	time_only	Δ (95% CI)
CWRU, P3	LR	0.457 [0.25, 0.75]	0.298 [0.15, 0.53]	+0.16 [−0.06, +0.40]
CWRU, P4	LR	0.681 [0.46, 0.84]	0.237 [0.08, 0.34]	+0.44 [+0.28, +0.61]*
Paderborn, P3	LR	0.745 [0.58, 0.86]	0.506 [0.37, 0.61]	+0.24 [+0.15, +0.33]*
Paderborn, P3b	LR	0.553 [0.31, 0.78]	0.381 [0.15, 0.57]	+0.17 [+0.06, +0.30]*
CWRU, P3	GBDT	0.505 [0.25, 0.85]	0.311 [0.15, 0.55]	+0.19 [−0.09, +0.47]
CWRU, P4	GBDT	0.748 [0.48, 0.91]	0.432 [0.13, 0.55]	+0.32 [+0.10, +0.61]*
Paderborn, P3	GBDT	0.700 [0.55, 0.81]	0.657 [0.51, 0.77]	+0.04 [−0.04, +0.11]
Paderborn, P3b	GBDT	0.587 [0.35, 0.81]	0.388 [0.19, 0.58]	+0.20 [+0.08, +0.35]*

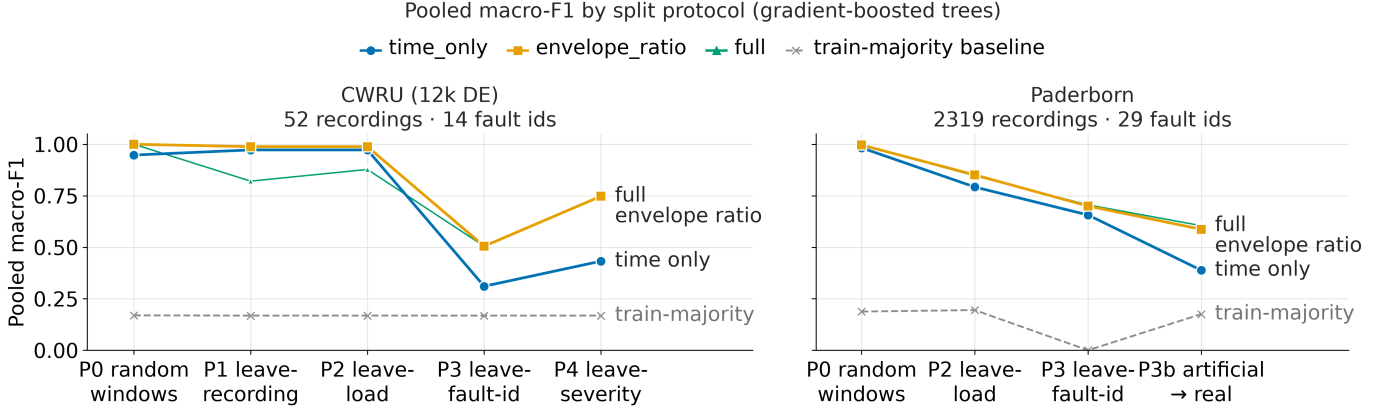


Figure 2. Pooled macro-F1 by split protocol with gradient-boosted trees: the same collapse from random windows to held-out faults as with logistic regression (Figure 1).

Table 4

SHAFT-SPEED SOURCE VERSUS ENVELOPE ADVANTAGE (LOGISTIC REGRESSION). “WITHIN 2%” IS THE SHARE OF WINDOWS WHOSE SPEED IS WITHIN 2% OF THE MEASURED VALUE; “USED” IS THE SHARE THAT KEEPS THE SPEED-DEPENDENT FEATURES (THE REST FALL BACK TO SPEED-FREE FEATURES); Δ AS IN TABLE 3.

Dataset	Speed source	Within 2%	Locked to 2×	Used	Δ envelope – time (95% CI)
CWRU	measured	100%	0%	100%	P3 +0.16 [−0.06, +0.40]; P4 +0.44 [+0.28, +0.61]
CWRU	estimated, rated $\pm 10\%$	100%	0%	100%	P3 +0.16 [−0.06, +0.40]; P4 +0.45 [+0.29, +0.62]
CWRU	estimated, 10–60 Hz	12%	67%	100%	P3 +0.20 [−0.05, +0.43]; P4 +0.39 [+0.22, +0.68]
Paderborn	measured	100%	0%	100%	P3 +0.24 [+0.15, +0.33]; P3b +0.17 [+0.06, +0.30]
Paderborn	estimated, rated $\pm 10\%$	57%	0%	37%	P3 +0.04 [−0.03, +0.11]; P3b −0.03 [−0.09, +0.02]
Paderborn	estimated, 10–60 Hz	38%	12%	40%	P3 +0.05 [−0.02, +0.12]; P3b +0.03 [−0.04, +0.09]

CWRU, leave-one-fault-identity-out (P3), pooled window counts; shade = row share

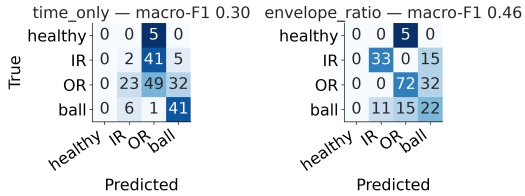


Figure 3. Pooled confusion matrices on CWRU P3 (leave-one-fault-out) for time_only (left) and envelope_ratio (right); counts are windows, and shading is the share of each true class.

VII. EDGE DEPLOYMENT

A. Sensor view and cost

We also pass each recording through a simulated low-cost sensor (Figure 10): a 6 kHz low-pass filter, resampling to

Table 5

COST PER FEATURE SET: FEATURES, FEATURE-VECTOR BYTES (FLOAT32), PARAMETERS (MODEL PLUS STANDARDISATION) AND HOST EXTRACTION TIME PER 4 s WINDOW (LAPTOP CPU, RELATIVE ONLY).

Set	Feat.	Bytes	LR par.	GBDT par.	ms
time_only	17	68	106	10,320	1.17
envelope_ratio	39	156	238	5,372	3.66
full	40	160	244	5,380	4.14

26.7 kSPS, and MEMS accelerometer noise. On Paderborn P3 this lowers time_only from 0.506 to 0.418, while envelope_ratio holds (0.745 to 0.756). Table 5 gives the model cost. The envelope logistic regression needs 238 parameters and a 156-byte feature vector.

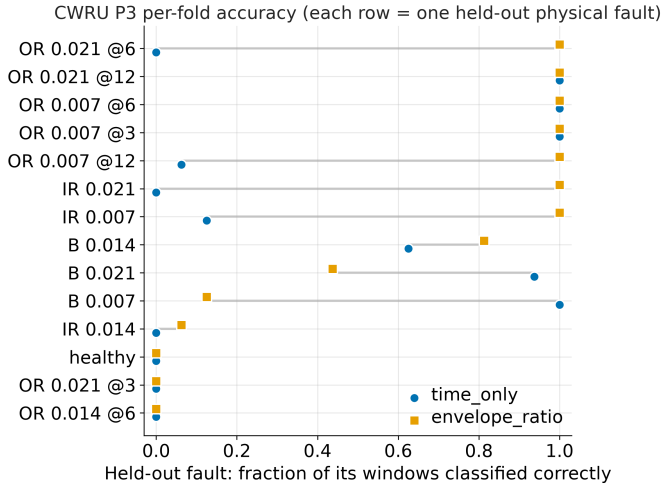


Figure 4. Per-fold accuracy on CWRU P3: each row is one held-out physical fault.

B. Edge gateway (*Orange Pi 5 Plus*)

We target a common factory layout: wired sensors stream raw vibration to one edge gateway that classifies for all of them. The gateway is an Orange Pi 5 Plus (Rockchip RK3588), with four low-power Cortex-A55 and four Cortex-A76 cores. Table 6 gives the measured timing. The Python pipeline takes 18.77 ms per 4 s window on one core and 6.99 ms for *time_only*. With all 8 cores busy it processes 177 windows/s, enough for about 353 sensors at a 2 s hop (645 for *time_only*).

We also timed the C++ port, verified against the Python implementation, on a single core. One Cortex-A55 core processes a 1.37 s window in 13.74 ms, $99\times$ faster than real time; one Cortex-A76 core takes 3.16 ms. Energy is estimated rather than measured, using a published wall-socket measurement of the board (about 5 W idle and 15 W at full load)^[12]. Dividing full-load board power by throughput gives about 84.97 mJ per window, or 42.48 mW per sensor at capacity.

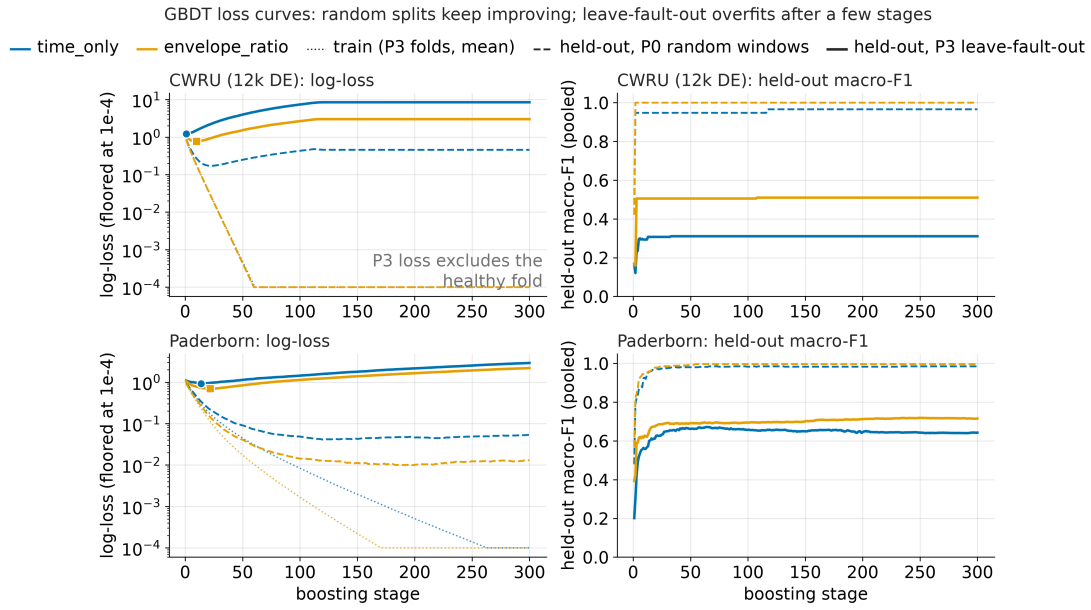


Figure 5. GBDT training and held-out log-loss (left) and held-out macro-F1 (right) per boosting stage, for random windows (P0, dashed) and leave-one-fault-out (P3, solid), CWRU (top) and Paderborn (bottom).

Table 6

ORANGE PI 5 PLUS EDGE GATEWAY. PYTHON: THE FULL PIPELINE PER 4 S WINDOW ON ONE CORE, AND WINDOWS PER SECOND WITH EVERY CORE BUSY; SENSORS SERVED IN REAL TIME AT A 2 S HOP (COMPUTE ONLY). C++: THE PORTED PIPELINE PER 1.37 S WINDOW PINNED TO ONE CORTEX-A55 (LITTLE) OR CORTEX-A76 (BIG) CORE. ENERGY: ESTIMATED, FULL-LOAD BOARD POWER DIVIDED BY THROUGHPUT.

Feature set	Python ms	Windows/s	Sensors	C++ A55 ms	C++ A76 ms	mJ/window (est.)
time_only	6.99	322	645	4.43	0.83	46.5
envelope_ratio	18.77	177	353	13.74	3.16	85.0

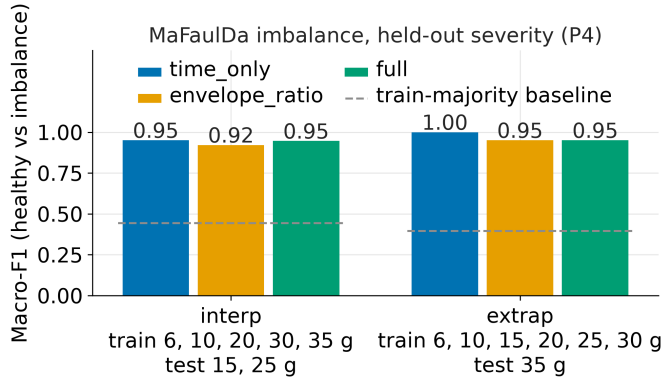


Figure 6. MaFaulDa healthy-versus-imbalance macro-F1 when the imbalance mass is held out: interpolation (train 6–35 g without 15 and 25 g, test 15 and 25 g) and extrapolation (test 35 g). The dashed lines are the train-majority baseline.

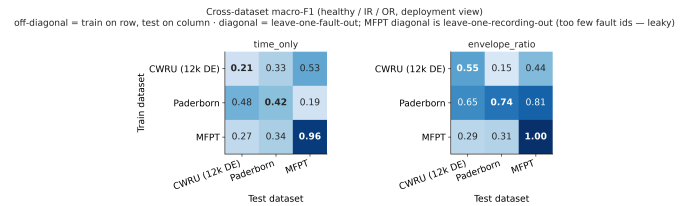


Figure 7. Cross-dataset macro-F1 (train on row, test on column; classes healthy, inner race, outer race) for time_only (left) and envelope_ratio (right). The diagonal is within-dataset leave-one-fault-out; the MFPT diagonal uses leave-one-recording-out and is inflated.

CWRU envelope spectra: IR peaks at BPFI, OR at BPFO (+ harmonics); the ball fault is weak

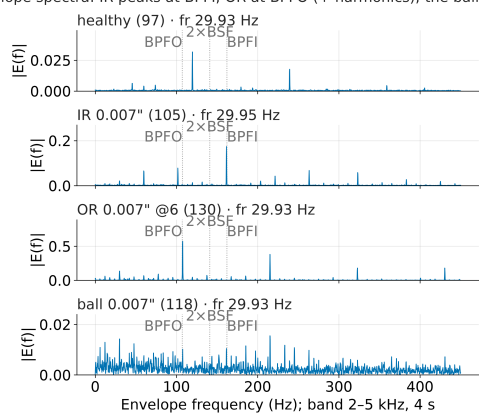


Figure 8. Envelope spectra of CWRU recordings (healthy, inner race, outer race, ball) with BPFO, BPFI and $2 \times \text{BSF}$ marked.

Estimated speed keeps the envelope advantage on CWRU (rated-speed prior) but loses most of it on Paderborn

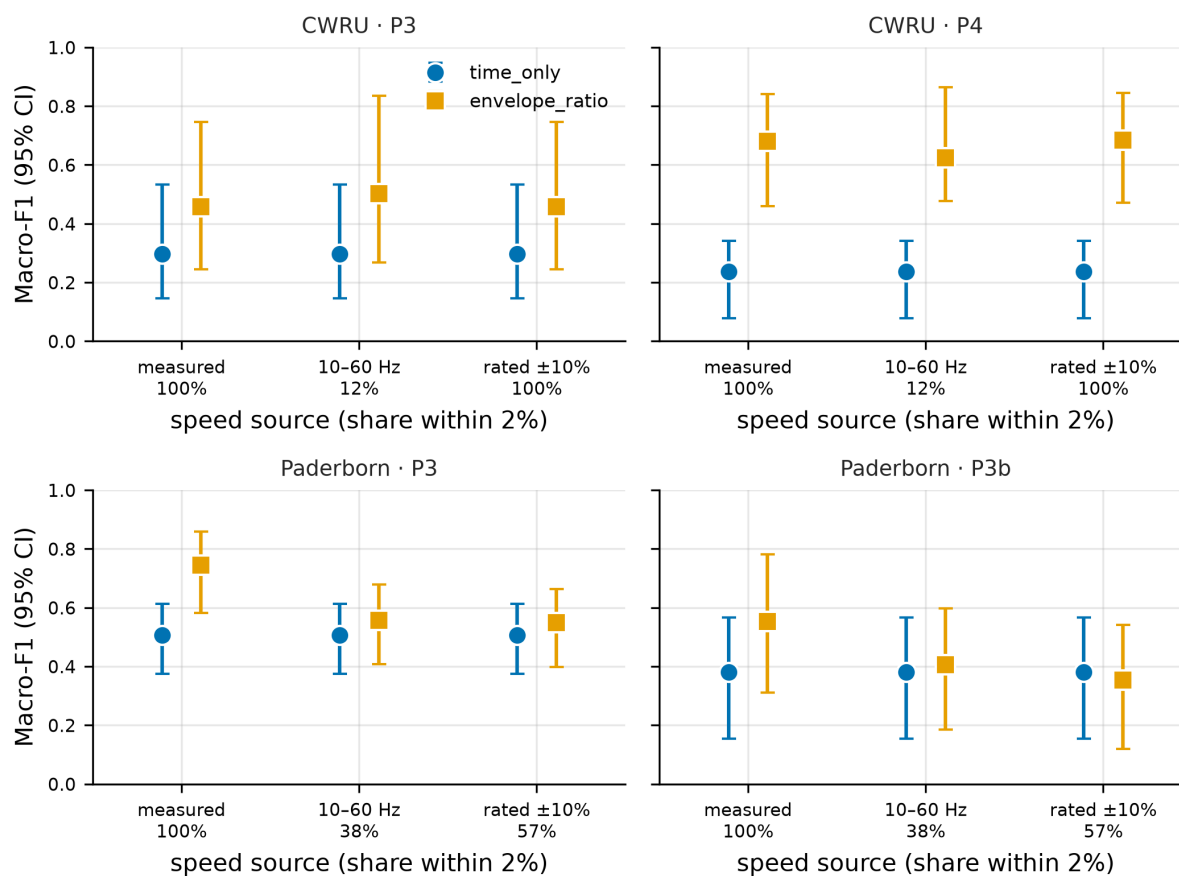


Figure 9. Honest-split macro-F1 with 95% fault-level confidence intervals for measured speed, speed estimated over 10–60 Hz, and speed estimated within $\pm 10\%$ of the rated speed.

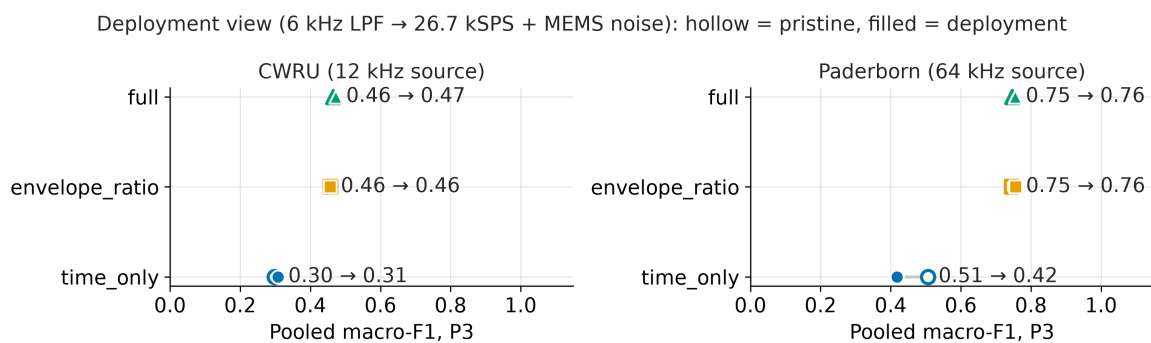


Figure 10. Pooled macro-F1 on P3 before (hollow) and after (filled) the simulated deployment sensor, for CWRU (left) and Paderborn (right).

VIII. LIMITATIONS

The intervals are wide: CWRU has 14 fault identities and Paderborn 29, and CWRU P3's gain is not significant. The envelope advantage requires measured or rated-speed-constrained speed, and on Paderborn it requires measured speed. CWRU's single healthy bearing cannot be tested under P3. Gateway energy is estimated from a published board figure, not measured. The sensors-served figure counts compute only, without network or I/O. The sensor view is simulated, and Paderborn's signal units are undocumented, so its added noise is approximate. MaFaulDa states no licence, and its cage faults are counted as outer race.

IX. CONCLUSION

Honest evaluation changes the picture of bearing fault detection. Random-window splits on CWRU score 1.000 for every feature set, whereas holding out the fault drops scores to 0.457–0.745. On those honest splits, envelope features at the kinematic fault frequencies outperform time-domain features. The advantage is significant on Paderborn leave-one-bearing-out, on artificial-to-real transfer and on CWRU leave-one-severity-out, but it depends on an accurate shaft speed. A rated-speed prior suffices on CWRU; Paderborn needs a tachometer or a better estimator. The envelope pipeline is cheap enough for a single low-power core of an edge gateway. Future work is measured rather than estimated energy, a microcontroller implementation for wireless sensors, and speed estimation where the shaft line is weak.

CODE AND DATA AVAILABILITY

All code, frozen split manifests, result files and figures are at https://github.com/AKSHAJ-SHELL/vibrometer_research. `make reproduce` reruns every experiment from the public datasets [1], [8], [10], [11].

AI-USE DISCLOSURE

[To be written by the author.]

REFERENCES

- [1] Case Western Reserve University Bearing Data Center, “Bearing data,” 2024. [Online]. Available: <https://engineering.case.edu/bearingdatacenter>
- [2] S. Zhang, S. Zhang, B. Wang, and T. G. Habetler, “Deep learning algorithms for bearing fault diagnostics—A comprehensive review,” *IEEE Access*, vol. 8, pp. 29857–29881, 2020, doi: 10.1109/ACCESS.2020.2972859.
- [3] D. Neupane and J. Seok, “Bearing fault detection and diagnosis using Case Western Reserve University dataset with deep learning approaches: A review,” *IEEE Access*, vol. 8, pp. 93155–93178, 2020, doi: 10.1109/ACCESS.2020.2990528.
- [4] S. Kapoor and A. Narayanan, “Leakage and the reproducibility crisis in machine-learning-based science,” *Patterns*, vol. 4, no. 9, Art. no. 100804, 2023, doi: 10.1016/j.patter.2023.100804.
- [5] W. A. Smith and R. B. Randall, “Rolling element bearing diagnostics using the Case Western Reserve University data: A benchmark study,” *Mech. Syst. Signal Process.*, vols. 64–65, pp. 100–131, 2015, doi: 10.1016/j.ymssp.2015.04.021.
- [6] J. Hendriks, P. Dumond, and D. A. Knox, “Towards better benchmarking using the CWRU bearing fault dataset,” *Mech. Syst. Signal Process.*, vol. 169, Art. no. 108732, 2022, doi: 10.1016/j.ymssp.2021.108732.
- [7] R. B. Randall and J. Antoni, “Rolling element bearing diagnostics—A tutorial,” *Mech. Syst. Signal Process.*, vol. 25, no. 2, pp. 485–520, 2011, doi: 10.1016/j.ymssp.2010.07.017.

- [8] C. Lessmeier, J. K. Kimotho, D. Zimmer, and W. Sextro, “Condition monitoring of bearing damage in electromechanical drive systems by using motor current signals of electric motors: A benchmark data set for data-driven classification,” in *Proc. Eur. Conf. PHM Soc.*, vol. 3, no. 1, 2016, doi: 10.36001/phme.2016.v3i1.1577.
- [9] J. Antoni, “The spectral kurtosis: A useful tool for characterising non-stationary signals,” *Mech. Syst. Signal Process.*, vol. 20, no. 2, pp. 282–307, 2006, doi: 10.1016/j.ymssp.2004.09.001.
- [10] Federal University of Rio de Janeiro (UFRJ), “MaFaulDa: Machinery fault database,” 2016. [Online]. Available: http://www02.smt.ufrj.br/~offshore/mfs/page_01.html
- [11] Society for Machinery Failure Prevention Technology, “MFPT bearing fault data sets,” mirror, 2025. [Online]. Available: <https://www.kaggle.com/datasets/emperorpein/mfpt-fault-datasets>
- [12] R. Carmo, “A review of the Orange Pi 5+,” *Tao of Mac*, Jan. 2024. [Online]. Available: <https://taoofmac.com/space/blog/2024/01/20/1800>